

FLEX-UF: Spatial Early Exit for Compute-Adaptive DCVC-UF Decoding

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Abstract

001 *A fast learned codec still spends the same synthesis depth*
002 *on every image region. We introduce FLEX-UF, a spatial*
003 *early-exit design for the intra decoder of DCVC-UF that*
004 *makes this depth a local decision. A shared latent and full-*
005 *frame stem feed a nested decoder; tiles leave through point-*
006 *wise adapters, while a shared repair and reconstruction*
007 *head restore a complete image. The design reuses compu-*
008 *tation across exits and preserves the learned representa-*
009 *tion. We study source-informed selection and learned rout-*
010 *ing against uniform depth and ordered dithering on 53 intra*
011 *frames at five quality indices. At a nominal 0.1 dB target,*
012 *the router removes 27.97% of modelled decoder multiply-*
013 *accumulates, including 2.54 percentage points beyond dither-*
014 *ing. A retrospective comparison under a common RGB-loss*
015 *cap against fine-tuned full-depth synthesis retains a 2.93-*
016 *point margin across all 265 frame-quality pairs. The ben-*
017 *efit contracts as the shallowest exit saturates, identifying*
018 *when spatial adaptation is useful. These results establish*
019 *an arithmetic-quality trade-off under source-calibrated con-*
020 *trol; complete bitstream latency remains to be measured.*
021 *A controlled 2/4/6/8/10/12-block study is specified to test*
022 *how shared early exits compare with independently trained*
023 *depth-reduced DCVC-UF codecs.*

024 1. Introduction

025 A reconstructed image contains regions with different de-
026 mands on a decoder. A smooth background and a textured
027 object pass through the same sequence of synthesis blocks,
028 even when the additional depth has different value in the two
029 regions. Modern learned codecs have made this sequence in-
030 creasingly fast. Their next efficiency opportunity is therefore
031 not only to execute each block faster, but to decide which
032 regions need to execute it at all.

033 We study this opportunity in the intra model supplied
034 with DCVC-UF [5]. Its synthesis transform offers a natural
035 place to vary computation: a stack of twelve constant-width

blocks between an upsampling stage and a reconstruction
head. Removing later blocks reduces synthesis work without
requiring a different coded latent. Yet simply truncating the
stack is insufficient. Intermediate features were not trained to
be final reconstructions; independent tile execution changes
boundary context; and deciding where to stop can consume
some of the work it saves. A useful early-exit codec must
address these three issues together.

FLEX-UF turns the synthesis stack into a nested spatial
decoder (Fig. 1). An initial full-frame stem preserves com-
mon context. Thereafter, each tile continues only as far as
its assigned depth. Exit-specific pointwise adapters make
intermediate features suitable for a common output space. A
lightweight repair on the stitched feature canvas and a shared
reconstruction head restore cross-tile communication near
the output. This arrangement concentrates adaptive execu-
tion in the expensive trunk, while retaining a single analysis
transform and entropy model.

The stopping rule is deliberately separated from the ex-
ecution mechanism. At the encoder, the source image can
supervise a search over reconstruction error. At the decoder,
a compact router can use already recovered features. The
former provides a useful decision reference but requires trial
reconstructions; the latter can avoid that acquisition cost
only when its control is calibrated without the test source.
We evaluate the current source-calibrated policy family and
make the remaining deployment experiment explicit. This
distinction lets the architecture be assessed without con-
fusing an inexpensive score prediction with an inexpensive
complete encoding procedure.

Our experiments ask how much of the saving comes from
early exit, and how much specifically comes from selecting
its spatial locations. Uniform depth measures the first effect.
Ordered dithering adds finer average costs without predict-
ing which region benefits from deeper synthesis. Learned
routing contributes a further 2.54 percentage points at the
0.1 dB target, on top of a 25.43% blind-mixture saving. That
contribution decreases near the ladder’s arithmetic ceiling.
The result is a useful design principle: content adaptation
matters most where several depths remain competitive, rather

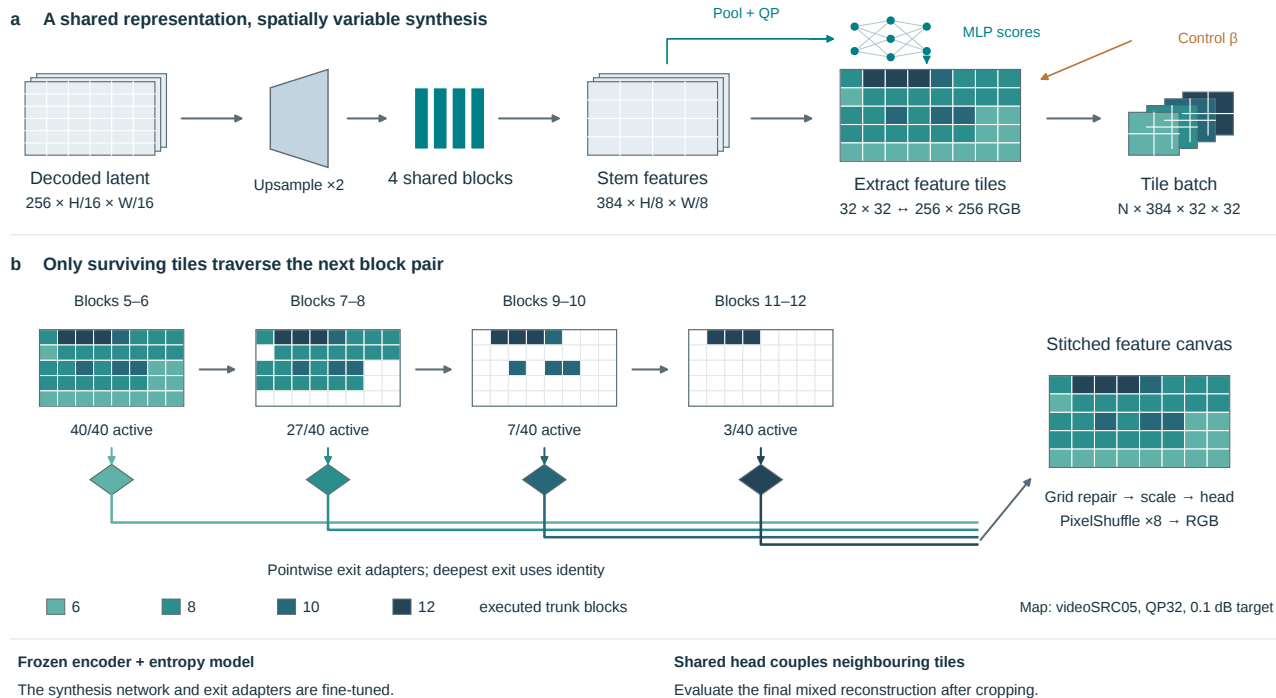


Figure 1. **FLEX-UF makes decoder depth a spatial decision.** **a**, A single decoded latent passes through four shared blocks before being partitioned into feature tiles. Stem statistics and QP feed the MLP; its scores and the source-calibrated control determine tile depth. **b**, Only surviving tiles traverse the next block pair. Exit adapters feed a stitched canvas, grid repair and a common head. The four reachable depths are 6/8/10/12; blank cells denote tiles that have already left. The displayed occupancy is recorded for videoSRC05, QP32, at a 0.1 dB target; tensor values are schematic. The independent 2/4/6/8/10/12 study in Sec. 5 is a separate control.

than where nearly all tiles can already take the cheapest exit.

This paper makes three contributions. First, it develops a DCVC-UF spatial early-exit path with shared-prefix execution, pointwise exit adaptation and full-frame boundary repair. Second, it evaluates the quality–arithmetic frontier using common reconstructions, strong spatially blind controls and paired uncertainty estimates. Third, it characterises where spatial adaptation adds value: the learned margin contracts as the shallowest exit saturates, while obtaining the stopping decision has its own encoder cost. The completed evidence is confined to DCVC-UF intra decoding. A controlled six-depth study is specified to separate shared-exit adaptation from independent capacity; its ongoing trainings and analytical scenarios are kept separate from measured reconstruction results.

2. Related Work

Efficient learned decoding. DCVC-RT argues that arithmetic count alone misses important operational costs, including memory traffic and execution overhead [3]. DCVC-UF advances throughput-oriented codec design [5]. Our intervention is confined to the synthesis transform of its intra model: the representation and entropy path are shared, and

computation varies spatially after entropy recovery. Shallow-decoder work offers a complementary approach by reducing synthesis capacity globally [11]. This motivates treating a fixed shallow decoder as a necessary control for an adaptive one, rather than attributing every reduction from full depth to successful routing.

Spatial adaptation and early exit. ClassSR assigns super-resolution patches to networks of different capacity [4], whereas APE predicts incremental benefit to select each patch’s exit depth [8]. RBQE adapts the depth used to enhance already compressed images [9]. These studies establish adaptive reconstruction effort. FLEX-UF applies this principle inside DCVC-UF synthesis, where a common coded representation, exit-feature alignment and tile-boundary context constrain the available stopping paths. Our evaluation also distinguishes the benefit of spatial placement from the cheaper average depth it enables.

Adaptive capacity within compression. Slimmable compressive autoencoders vary model width [10]; AdaNIC varies transform width spatially using blockwise operators and a learned router [7]; cgSlimDecoder controls decoder channel

widths under complexity constraints [2]. These methods already demonstrate adaptive computation in learned compression. Our focus is nested depth in DCVC-UF, executed through a shared stem and a shrinking tile batch. We do not claim that content-dependent codec computation itself is new.

Complete-codec selection. Spatial competition selects specialized codecs by region and communicates a mode map [1]. It is the closest comparison for our planned independent depth bank, where each expert owns its analysis and entropy parameters. That system differs from shared-latent early exit: choosing an expert before encoding changes both its representation and rate. The six-depth control therefore tests a separate capacity frontier rather than retroactively adding shallow exits to the present checkpoint.

3. FLEX-UF: Spatial Early Exit

3.1. A shared representation, a variable synthesis depth

Let $\hat{y}$ be the recovered latent of the fixed DCVC-UF analysis transform and entropy model. The synthesis network upsamples it by two and processes 384-channel features at one eighth of the RGB resolution. We divide its twelve trunk blocks into six consecutive pairs. A split index determines which pairs run on the full frame and which run conditionally on tiles. In the evaluated configuration, four blocks form the common stem and the remaining eight form a four-stage conditional suffix.

The stem output is partitioned into 32×32 feature tiles, each covering 256×256 RGB pixels. For a tile assigned depth k , let $f_t^{(k)}$ be its feature after the corresponding block pair and let A_k be its adapter. The complete image is reconstructed as

$$\hat{x}_k = H\left(q \odot G\left[\text{stitch}_t A_{k_t}(f_t^{(k_t)})\right]\right), \quad (1)$$

where G is grid-conditioned repair, q is the decoder’s quality scaling, and H is the common output head, including pixel shuffle. Adaptation acts on features before the shared head, so tiles with different stopping depths need not be stitched directly in RGB space.

Reachable depth and actual execution. Although the configuration contains six nominal exit indices, the first two precede the split and are clamped in this evaluation. The distinct reachable depths are $\mathcal{K} = \{6, 8, 10, 12\}$. At each conditional stage the implementation gathers only the surviving tiles, executes the next block pair, and writes departing tiles through their adapters. A deeper tile reuses all computation already performed. Evaluating all four candidates and masking their RGB outputs would not implement this

execution saving. For the recorded example in Fig. 1, the active population decreases from 40 to 27, seven and three tiles.

3.2. Exit adaptation and boundary consistency

An intermediate feature is not automatically compatible with the final head. The exit adapter supplies the missing transformation with pointwise residual operators. Its residual output is zero-initialised, preserving the inherited feature at initialisation. The evaluated depth-scaled adapters allocate capacity according to the depth they replace. Pointwise operators avoid introducing a new spatial receptive field at each exit; their arithmetic is included in the per-exit cost model. The deepest exit uses an identity connection rather than an additional learned adapter.

Independent tile execution still removes context from the trunk’s spatial operators. The conditional trunk uses replicate padding without a halo. After stitching, a depthwise spatial filter and pointwise mixing correct the full feature canvas. A learned gate indexed by position within the tile gives the repair access to the known boundary lattice. The head then operates on the complete canvas. This design spends a shared output-stage cost to recover boundary context instead of expanding every conditional block with overlapping halos. Whether that trade is favourable is an empirical question: repair removal, halo size and patch size belong in separate matched ablations, with retraining distinguished from an inference-only intervention.

3.3. Training the exits and the stopping predictor

The shared-latent model is initialised from released DCVC-UF weights. Analysis and entropy parameters remain frozen; the synthesis network, adapters and repair are fine-tuned. Training samples tile depths and applies reconstruction supervision through the mixed tiled path. Thus the adapters learn under the same stitching and full-frame head used at inference. The training implementation also supports a full-depth reconstruction anchor against the frozen release and adjacent-exit feature distillation. The inspected e15 training configuration enables both terms. Its objective can be written as

$$\mathcal{L} = \mathbb{E}_k[R + \lambda D(x, \hat{x}_k)] + a\mathcal{L}_{\text{anchor}} + b\mathcal{L}_{\text{feature}}. \quad (2)$$

Here the latent-rate term R is shared across exits. With the front end frozen it does not optimise exit-dependent rate. The archived fine-tuning configuration uses 512-pixel crops and batch size eight; it is distinct from the new full 105-epoch official-recipe depth study in Sec. 5. The supplement records the available training provenance and its limits.

The evaluated stopping predictor uses a 384-to-48 pointwise stem projection, per-tile mean and standard deviation, and the normalised QP. A two-layer 256-unit MLP predicts exit scores. Inspection of the actual router checkpoint shows

that its live inputs are *stem* and *QP*; the more general implementation contains latent, entropy-scale and bit-feature branches, but these do not supply content information in this run. This distinction is important for reproducing the model rather than merely its class name.

For tile errors $M_{t,k}$ and normalised costs c_k , the training target is softened from the Lagrangian $L_{t,k} = M_{t,k} + \lambda_c c_k$:

$$p_{t,k}^* \propto \exp \left[-\frac{L_{t,k} - \min_j L_{t,j}}{\tau \max(\max_j L_{t,j} - \min_j L_{t,j}, \epsilon)} \right]. \quad (3)$$

The checkpoint records $\lambda_c = 1.3 \times 10^{-5}$ and $\tau = 0.1$. Soft-label cross-entropy, together with the implementation's regret term, reduces the penalty for confusing nearly equivalent exits. This loss trains an allocation predictor; it is not a calibrated bound on final image error.

3.4. Stopping under a quality target

For equal padded tiles, the arithmetic model is

$$C(\mathbf{k}) = \frac{1}{N} \sum_t c_{k_t}, \quad S(\mathbf{k}) = 100[1 - C(\mathbf{k})]. \quad (4)$$

Shared stem and output work, exit adapters and conditional depth are included in c_k . Router execution, signalling and general system overhead are outside this decoder MAC quantity. The four costs are approximately $\{0.609, 0.758, 0.870, 1.010\}$ relative to full-frame synthesis. In particular, a tiled all-deep map can cost more than the full-frame reference because repair is still executed.

Source-informed search minimises $M_{t,k} + \lambda c_k$ per tile and adjusts λ for a target table loss. The learned alternative uses

$$k_t(\beta) = \arg \max_k \{\ell_{t,k} - \beta c_k\}, \quad (5)$$

where ℓ denotes log scores. Uniform depth selects one exit per frame. Bayer dithering mixes adjacent exits with a fixed spatial ordering. Its mixture fraction is calibrated under the same target, so it controls for finer average depth without predicting spatial difficulty.

In the recorded sweep, all controls are calibrated per source frame using stored error tables. This is an attainable allocation study, not a test of autonomous held-out calibration. Increasing β cannot increase selected cost, but source distortion need not be monotone; the supplement audits the finite assignment path for both router and dither.

3.5. Quality references and decision information

The table target compares padded RGB candidate errors to a released-weight full-depth reference. Final reported loss instead compares the cropped mixed output to the cropped, fine-tuned e15 full-frame output. These anchors are not interchangeable: the encoder/entropy weights match, while inherited synthesis weights have changed. Full-frame repair

also couples neighbouring tiles. A table threshold therefore does not guarantee the same numerical loss after final reconstruction. We retain this distinction in the nominal sweep and add a retrospective common-reference delivered-loss comparison.

The information path also determines the deployment cost. Encoder-selected maps require side information. A source-calibrated router needs its control parameter communicated or reproducibly recovered. A decoder-local router with a fixed validation-selected control can avoid a per-frame source table, but its quality–cost frontier must be measured separately. Common latents establish a shared payload opportunity; they do not make unmeasured routing headers or complete bitstreams free.

4. Experiments

4.1. Evaluation protocol

We evaluate the first intra frame of 53 CTC sequences at QP values $\{0, 16, 32, 48, 63\}$, giving 265 frame–QP pairs. Every policy uses the same pinned e15 shared-latent checkpoint, the same tile geometry and the same per-exit cost vector. The learned rule uses one fixed router checkpoint. Six target budgets span 0.05–0.5 dB. The archive contains actual mixed reconstructions; the present analyses reuse those outputs without generating new GPU timings. All reported numerical decoder comparisons concern DCVC-UF.

Two complementary protocols answer different questions. The *nominal sweep* compares maps selected under the same table target, retaining the common feasible cases of all four rules: 204 pairs at 0.05 dB, 263 at 0.1 dB and 265 at each larger target. The *delivered-cap analysis* reselects from the outputs already measured at those targets under a common cropped RGB-loss cap, with the full-frame e15 output as an explicit fallback. It retains all 265 pairs at every cap. The second protocol is retrospective and source-aware; its cost omits acquiring and rejecting candidates.

Confidence intervals use 5,000 paired bootstrap draws over sequences, keeping QPs from the same sequence together. They quantify content variation at fixed weights. They do not replace independent training seeds. A separate fixed-cohort comparison retains the same 204 pairs across the entire nominal sweep, avoiding changes of sample membership as the target relaxes.

4.2. The reconstruction value of depth

The stored uniform-exit reconstructions provide a direct depth control. Across all 265 pairs, their mean padded RGB losses against the released full-frame reference are 0.205/0.097/0.061/0.038 dB at 6/8/10/12 blocks. Their modelled MAC savings are 39.12/24.22/13.03/–0.95%, respectively. Additional depth has diminishing average benefit and non-uniform local benefit: mean within-frame tile-gain

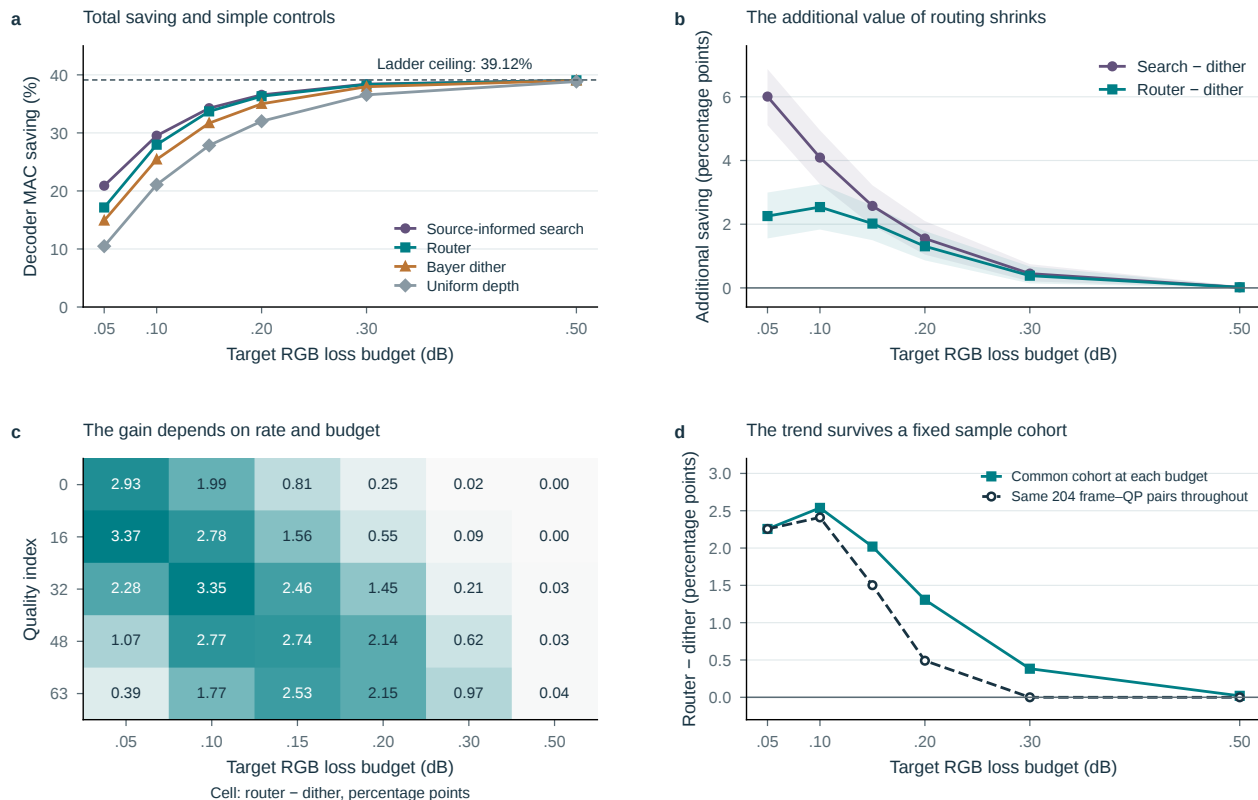


Figure 2. **Early exit supplies the saving; spatial routing adds a budget-dependent margin.** **a**, Modelled decoder MAC reduction for four stopping rules. **b**, Paired advantages over Bayer dithering, with 95% sequence-cluster intervals. **c**, Router-dither difference by QP and target. **d**, Repeating the comparison on the same 204 pairs retains the trend. All curves use per-source calibration. MAC savings exclude router and signalling overhead and are not measured runtime reductions.

IQRs are 0.075/0.034/0.024 dB for successive block pairs. The supplement gives the distributions and paired intervals. These measurements concern actual uniform maps on padded RGB; they motivate spatial allocation but are not cropped mixed-map losses or results for independently trained shallow codecs.

4.3. Early exit reduces work; routing improves its allocation

At the nominal 0.1 dB target, FLEX-UF with learned routing saves 27.97% of modelled decoder MACs. Source-informed search reaches 29.52%, while uniform depth and dithering reach 21.07% and 25.43%, respectively (Fig. 2). The ordering separates three effects. Shallower synthesis supplies most of the reduction. Mixing adjacent depths improves the available cost resolution. Content-dependent placement contributes a further 2.54 percentage points over dithering, with a paired 95% interval of [1.83,3.26].

The total reduction quantifies the early-exit mechanism; the increment over dithering quantifies learned allocation.

Reporting both prevents the stopping predictor from receiving credit for all the work removed by cheaper depth.

4.4. The gain persists under a common delivered-loss cap

The nominal target is an imperfect proxy for final quality because of the reference and support differences described in Sec. 3. We therefore ask a narrower question using the reconstructions already in the archive: does the relative saving persist when all selected outputs must satisfy the same measured RGB-loss cap?

For each policy and frame-QP pair, we deduplicate the maps measured at the six nominal targets, retain those meeting the cap within 10^{-4} dB, and choose the cheapest. The evaluated full-frame e15 output is available with zero relative loss and zero saving. No synthetic output or interpolated quality point is introduced. Repeated maps have identical recorded RGB losses and MAC costs, which we check before deduplication.

Table 1 reports the 0.1 dB case. The router saves 27.75%

Table 1. **Recorded outputs under a common 0.1 dB RGB-loss cap.** All 265 pairs are included. Loss is relative to cropped e15 full-frame synthesis. MAC saving excludes the retrospective candidate-selection cost.

Policy	MAC saved (%)	Mean loss (dB)	Fallbacks (/265)
Uniform depth	22.98	0.0670	11
Bayer dither	24.81	0.0791	2
Learned router	27.75	0.0786	1
Source search	29.19	0.0779	2

versus 24.81% for dithering, a 2.93-point paired difference [2.11,3.79]. Their mean achieved losses are also close: 0.0786 and 0.0791 dB. Only one router case uses the full-frame fallback, compared with two dither cases. The retained margin supports the qualitative conclusion from the nominal sweep. It does not establish a deployable guarantee: selecting after final quality is known still requires source access, and the finite candidate pool is not a global optimum.

4.5. Why the routing margin contracts

The cheapest reachable depth imposes a 39.12% ceiling on the arithmetic model. As the target relaxes, more tiles can take that exit and there are fewer consequential spatial decisions. The nominal router–dither margin declines from 2.54 points at 0.1 dB to 1.31 at 0.2 dB, 0.38 at 0.3 dB and 0.02 at 0.5 dB. Under the delivered-cap protocol the margin similarly falls from 2.93 to 0.35 points between 0.1 and 0.3 dB. The fixed-cohort result and QP breakdown in Fig. 2 indicate that this behaviour is not explained solely by a changing feasible subset. At the 0.1 dB delivered cap, routing saves more in 170 pairs, less in 24, and ties in 71. At 0.5 dB, 262 of 265 pairs tie; the residual gain comes from one sequence.

The maps in Fig. 4 make the mechanism visible. At the tighter target, the router spends different depths on different regions; under the relaxed target, the maps largely converge to the cheapest depth. The example is chosen from the five archived thumbnails by proximity to the aggregate 0.1 dB routing margin. Its source luma gives spatial context; it is not a reconstruction crop or evidence of perceptual equivalence.

A histogram-preserving diagnostic further separates spatial placement from exit proportions. Randomly permuting a map leaves its cost unchanged. We compute the exact expected table MSE under that permutation and compare it with the recorded assignment. At 0.1 dB, the router’s mean placement gain is 0.0213 dB [0.0167,0.0262], whereas dithering’s is 0.0007 dB [−0.0002,0.0017]. This indicates useful spatial association conditional on the chosen histogram. The calculation uses the source-error table; replaying permuted maps through repair and the final head is a separate ablation.

4.6. What the encoder pays for a stopping decision

Source-informed search needs candidate reconstructions to populate its error table. The archived stage benchmark recorded 1,347.65 ms for six trials, 1,023.26 ms after deduplicating the four distinct reachable outputs, and 336.87 ms using a shared tiled pass. Reusing the common prefix therefore reduces this recorded table-construction statistic by about fourfold. The values are upper medians of 24 pooled trials over eight frames at QP32; raw repetition distributions and contemporaneous device load were not retained. They describe information acquisition, not an end-to-end codec speedup. The supplement visualises these stages and their accounting.

A fixed-control predictor can avoid this table, but the evaluated source-calibrated router cannot: its per-frame β uses the same error information. The next latency experiment must compare full decision paths, including feature generation, calibration, signalling, conditional execution and final verification. At a loose quality budget, where the router and blind maps nearly coincide, even small additional overhead may remove the practical benefit of content prediction.

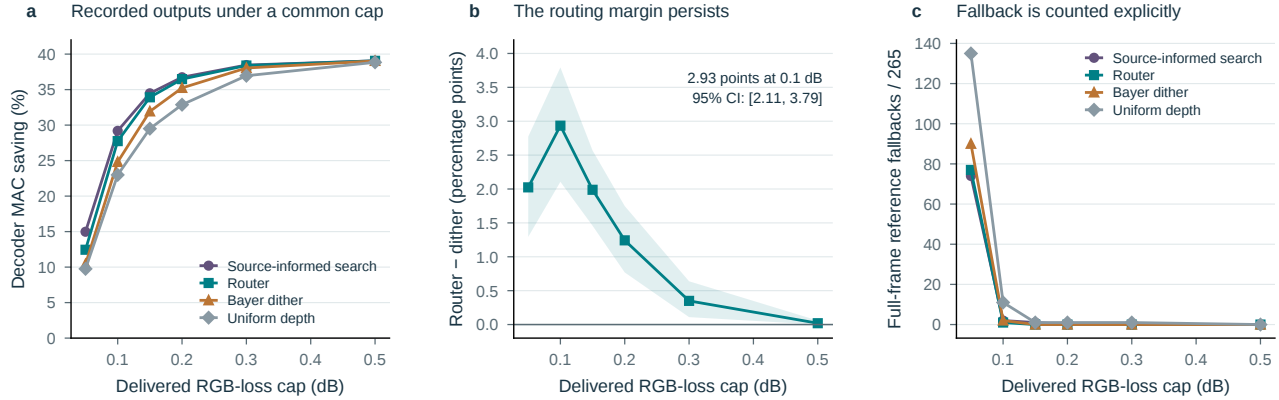
5. A Controlled Six-Depth Study

Shared early exit trades representation reuse against specialised capacity. Our next study compares independently trained DCVC-UF codecs retaining 2/4/6/8/10/12 synthesis blocks. Their encoder and entropy parameters also learn, so their latents are not interchangeable exits of the current model. Table 2 reports exact capacity and experimental status; completed matched RD and latency results are pending.

Each variant preserves the opening upsampler and output head. D2 removes 73.0% of D12’s synthesis parameters but only 24.6% of complete-codec parameters, because analysis and entropy capacity remain. A same-recipe D12 trained from scratch must complement the released anchor to isolate depth from training provenance. Likewise, attributing a gap between this bank and the older e15 fine-tuning run specifically to weight sharing requires matched data, initialisation and optimisation controls.

Training and evaluation. The independent runs retain the pinned official 105-epoch recipe [6], all 384,795 Open Images train_0/1/2 images, batch size 16 and FP32. The supplement records the schedule and execution settings. Four validation crops monitor health; final comparison requires common achieved RD, complete coded bytes and paired timing. Equal QP is not equal rate across codecs.

Later encoder-side selection. The proposed MLP in Fig. 5 selects an expert before patch encoding and transmits its identity. Controls are full-frame D12, patchwise D12, a



265 frame-QP pairs throughout · fine-tuned e15 anchor · retrospective source-aware candidate selection

Figure 3. **The spatial-routing margin survives a common measured quality cap.** **a**, Lowest-cost eligible output within each policy’s recorded candidate pool. **b**, Paired router–dither saving and 95% sequence-cluster interval. **c**, Full-frame fallback counts. All 265 pairs remain in every comparison. This is retrospective selection among actual reconstructions, with fine-tuned e15 as the common anchor; candidate-acquisition time and released-reference quality are not measured.

a Spatial decisions converge as the quality budget relaxes

videoSRC05_1920x1080_25.yuv · QP32 · archived source luma

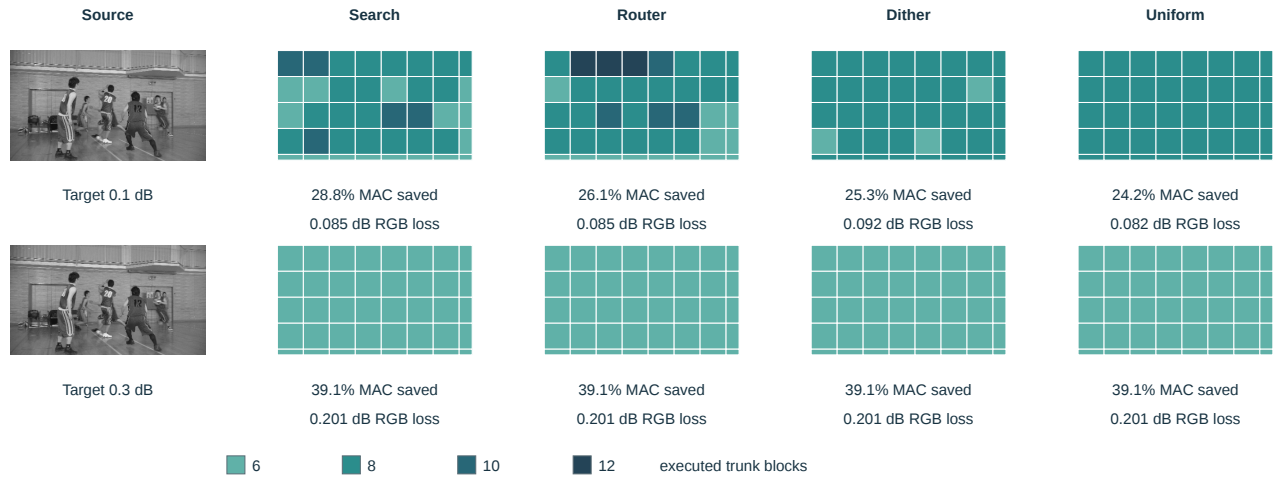


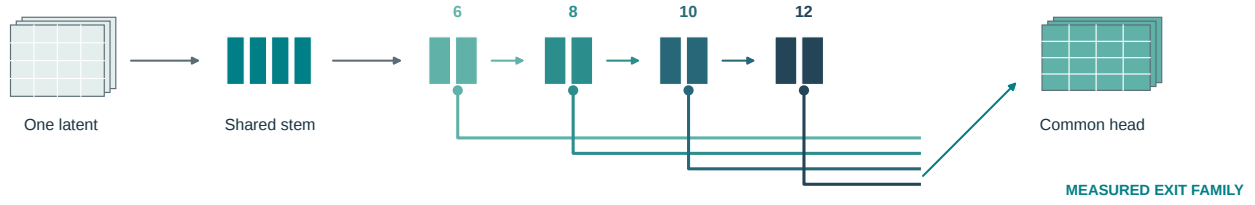
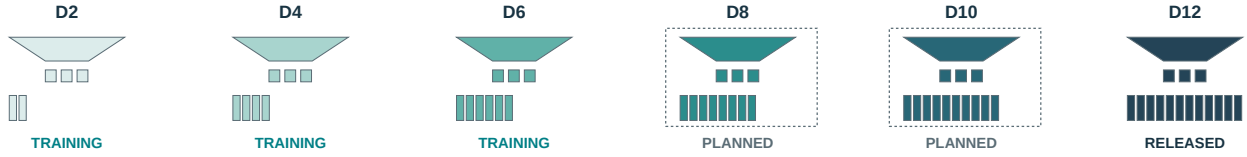
Figure 4. **Relaxing the quality budget removes the need for spatial differentiation.** Recorded exit maps at two targets on the same QP32 frame. Colour indicates total executed trunk depth; annotations give MAC saving and measured RGB loss. Source luma is shown only for spatial context. Maps are cropped to the valid image extent, and the example-selection rule is stated in the text.

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validation-selected fixed expert and blind mixtures. Leave-one-expert-out tests measure patch-specific value. Training, calibration and test images stay separate; timing includes grouping, entropy resets, signalling and fallback. The supplement’s dashed runtime scenarios expose assumed fixed

costs without inventing quality or speed measurements for unfinished models.

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a Shared early exit: one representation, nested synthesis**b Independent depth controls: each codec learns its own representation**

Same-recipe D12 is an additional control. The released anchor alone does not isolate depth.

c Later routing study: select before encoding; signal which expert owns the payload

Figure 5. **The six-depth study separates early exit, model capacity and later routing.** **a**, The completed shared-latent experiment reuses one representation and nested synthesis. **b**, The planned capacity ladder trains complete DCVC-UF codecs with different retained synthesis depths; each owns its analysis and entropy parameters. **c**, A subsequent encoder-side selector can group patches by expert and transmit expert identity with the coded payload. D2/D4/D6 trainings are ongoing; D8/D10 and the routing stage are prospective. Geometry is schematic and encodes no predicted quality or throughput. A same-recipe D12 control complements the released reference.

Table 2. **DCVC-UF capacity controls, not quality predictions.** Parameter counts are architectural. Final matched RD and timing are pending for the controlled depth experiment. All figures are in millions.

Depth	Decoder	Complete codec	Status
D2	3.84	31.79	Training
D4	5.92	33.87	Training
D6	8.00	35.95	Training
D8	10.08	38.02	Planned
D10	12.15	40.10	Planned
D12	14.23	42.18	Released anchor

6. Discussion

FLEX-UF exposes two operating regimes. Tight quality constraints leave room for useful learned placement. Near the shallowest-exit ceiling, blind mixtures approach the same arithmetic cost, and routing overhead can outweigh the remaining margin. A practical stopping policy should therefore be chosen jointly with the quality budget and measured device cost.

The principal limitation is the gap between this allocation evidence and a complete codec benchmark. The current controls use source calibration, reported losses use the fine-tuned full-depth anchor, and MAC counts exclude routing and signalling. A common-reference quality replay and a held-out control sweep are required before a practical guarantee can be claimed. Actual bytes and paired wall-clock timings must then include entropy work, packing, tile movement and fallbacks. Perceptual metrics and boundary crops also remain necessary; a small RGB PSNR loss alone does not prove visual indistinguishability.

7. Conclusion

We presented FLEX-UF, a DCVC-UF spatial early-exit design that combines a shared synthesis stem, conditional tile depth, pointwise adaptation and full-frame repair. Recorded reconstructions show substantial decoder arithmetic reductions and a further, budget-dependent gain from learned placement over dithering. A common delivered-loss analysis preserves that trend. Controlled depth and complete-bitstream experiments are the next tests of whether these arithmetic gains produce a practical codec speedup.

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